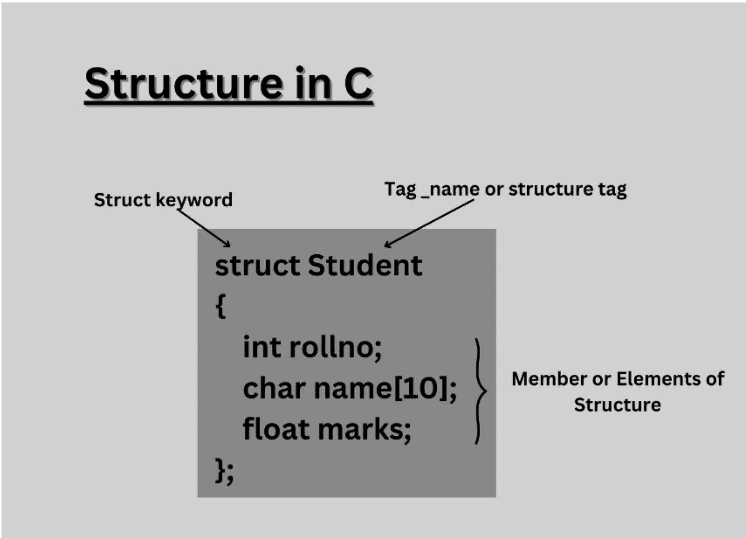
CPP NOTES -DAY 23

STRUCTURES

A struct in C++ is a user-defined data type that allows grouping of variables of different types under a single name. This is particularly useful for organizing complex data structures. Unlike arrays, which can only hold elements of the same type, a struct can contain variables of different types, such as integers, strings, and Booleans



Different formats:

1. *Struct s{….}v1,v2,v3;*
2. *Struct s{…};*
   Struct s v1;

Summary of Differences

Feature	Case 1 (struct s { ... } v1, v2, v3;)	Case 2 (struct s { ... }; struct s v1;)
Definition & Declaration	Defines the struct type and declares multiple variables in one line.	Defines the struct type and declares variables in separate lines.
Number of Variables	Declares multiple variables of the struct type in one statement.	Declares one variable at a time (though you can declare more later).
Code Structure	More concise for declaring multiple variables.	More separated, allowing flexibility in declaring variables.
Typical Usage	Convenient for declaring multiple variables of the same type.	Suitable when you want to define the struct first, and then declare variables when needed.

3 parts of structure:

1. Defenition
2. Declaration
3. Use members

2 types of operator:

1. Dot(.) – normal variable
2. Arrow(-->) – when declared as pointer

To access: nameOfVariable.memName(v1.name)

FYI:

- *Valgrind is used to identify memory leaks*
- *With new alloc we should use delete method to free the memory and in new casting is not needed.*
- *If you call free(ptr) twice on the same pointer in C (known as a double free), you invoke undefined behavior.*
- *Purpose of struct: related items are bunched together*